

# Qualitative Inorganic Salt Analysis Manual

## Complete Practical & Viva-Voce Guide (Group 0 to VI)

Chapter / Focus Area: Class 11 & 12 Laboratory Practical Chemistry

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### 1. Systematic Anion Identification

- Dilute H<sub>2</sub>SO<sub>4</sub> Group: Carbonate (CO<sub>2</sub> effervescence with lime water), Sulphide (H<sub>2</sub>S rotten egg smell), Acetate (vinegar smell).
  - Concentrated H<sub>2</sub>SO<sub>4</sub> Group: Chloride (pungent HCl gas, white fumes with NH<sub>4</sub>OH), Nitrate (Brown Ring Test [Fe(H<sub>2</sub>O)<sub>5</sub>(NO)]<sup>2+</sup>).
  - Independent Group: Sulphate (BaCl<sub>2</sub> → White BaSO<sub>4</sub> ppt insoluble in conc. HCl).
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### 2. Systematic Cation Separation Scheme

- Zero Group: NH<sub>4</sub><sup>+</sup> + NaOH → NH<sub>3</sub> gas (turns red litmus blue, Nessler's reagent → brown ppt).
  - Group I: Pb<sup>2+</sup> + Dil. HCl → White PbCl<sub>2</sub> precipitate (soluble in hot water).
  - Group II: Cu<sup>2+</sup> + H<sub>2</sub>S in acidic medium → Black CuS precipitate.
  - Group III: Fe<sup>3+</sup>, Al<sup>3+</sup> + NH<sub>4</sub>Cl + NH<sub>4</sub>OH → Reddish-brown Fe(OH)<sub>3</sub> or gelatinous white Al(OH)<sub>3</sub>.
  - Group IV: Zn<sup>2+</sup>, Ni<sup>2+</sup>, Mn<sup>2+</sup> + H<sub>2</sub>S in ammoniacal medium → ZnS (dirty white), NiS (black).
  - Group V: Ba<sup>2+</sup>, Sr<sup>2+</sup>, Ca<sup>2+</sup> + (NH<sub>4</sub>)<sub>2</sub>CO<sub>3</sub> → White carbonate precipitates (flame test confirmation).
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